

Sampling Bias & Poor-Quality Data

Why having a large dataset isn't enough: Understanding bias, data quality, and model reliability in Machine Learning.

1. Literary Digest-ի Հայտնի Միսալը

The 1936 Presidential Election Blunder

Massive Scale

10 million questionnaires sent out, with ~2.4 million responses received. Predicted: Landon (57%).

Real Outcome

Actual result: Roosevelt won with 62%. Huge dataset did not prevent a completely incorrect prediction!

2. Sampling Bias

Why Did the Method Fail?



Source Lists: Addresses were pulled from telephone directories, magazine subscriber lists, and club memberships.



Wealth Representation: These lists overrepresented wealthy individuals who heavily favored Republicans at the time.



Sample $\neq$ Population: Because the sample did not match the actual population demographics, severe bias occurred.

3. Nonresponse Bias

The Silence of the Majority

The 75% Gap

Out of 10 million surveys, about 75% did not respond. Are non-respondents' opinions identical to respondents?

Key Distinction

Sampling Bias: Bias introduced during sample selection.
Nonresponse Bias: Bias because specific groups choose not to respond.

4. Funk Music YouTube Example

Search Engine & Geographic Bias

Popularity Trap

Training on YouTube search results favors high-view, popular artists, overrepresenting them in the training data.

Geographic Shift

Searching "funk music" in Brazil yields Funk Carioca, which differs vastly from James Brown's funk, skewing model behavior.

5. Poor-Quality Data

Correct Population, Bad Measurements

- ✗ **Sampling Bias:** Choosing the wrong data subset.
- ⚠ **Poor-Quality Data:** Collecting the right people/items, but the data features errors, outliers, noise, or missing values.

6. Outliers & Missing Values

Dealing with Data Anomalies

Anomaly Type	Example	Resolution Strategy
Outlier	Age: 250 (Data entry error)	Delete the outlier or correct the invalid value.
Missing Values	Customer Age: ? (Blank fields)	Remove feature/rows, impute with median, or test models with/without.

The Three Levels of Data Failure

1. Insufficient Data

Too few samples (e.g., 100 samples).

2. Nonrepresentative

Enough data, but from a biased demographic group.

3. Poor-Quality

Right population, but corrupted with noise/errors.